

MpVH: Mamba Dynamically Prompted Vision Transformer for Deep Hashing

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Abstract

Deep hashing has demonstrated strong computational efficiency for large-scale image retrieval. However, existing deep hashing methods primarily focus on optimizing the distribution of hash codes in the Hamming space, while paying insufficient attention to the loss of rich descriptor information caused by compression and quantization during hash mapping. To address this, we propose a Mamba dynamically prompted Vision Transformer for hashing-based image retrieval. MpVH exploits the complementary sensitivities of ViT and Mamba to visual content: ViT captures global contextual semantics, whereas Mamba is more responsive to locality-biased discriminative information through selective state-space modeling and multi-path spatial scanning. Based on this observation, we introduce a Mamba prompter to generate locally discriminative prompts and a dynamic mixture-of-experts adapter to aggregate historical prompts and bridge the semantic gap across architectures. These prompts guide the ViT backbone to learn importance-aware image descriptors, preserving more discriminative information during feature compression. Extensive experiments on three benchmark datasets demonstrate that MpVH achieves competitive retrieval accuracy while maintaining high encoding efficiency.

CCS Concepts

• Computing methodologies → Visual content-based indexing and retrieval.

Keywords

Deep hashing; Vision Transformer; Mamba; Mixture-of-experts

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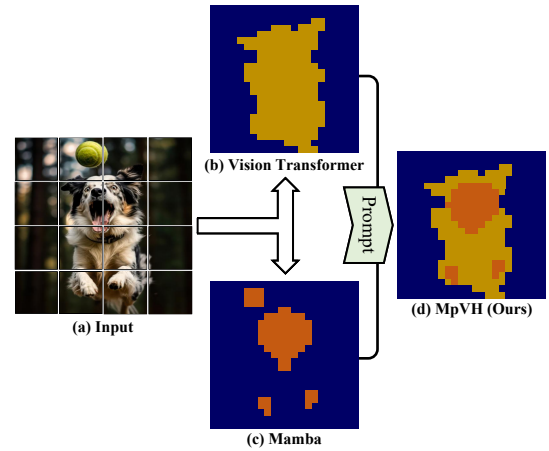


Figure 1: Comparison among different methods.

1 Introduction

In recent years, with the rapid growth of visual information, traditional image retrieval methods have encountered bottlenecks in computational complexity and storage cost, making it difficult to satisfy the practical requirements of efficient retrieval. In contrast, hashing-based image retrieval methods [27, 33] using compact binary encoding have attracted considerable attention for large-scale image processing tasks due to their high computational efficiency and low storage overhead. These methods aim to map high-dimensional image descriptors into compact binary codes via a hashing function, ensuring that semantically similar samples are assigned similar hash codes, while maximizing the Hamming distance between dissimilar ones. From this perspective, the discriminability of the learned hash codes depends heavily on the quality of the underlying image descriptors.

With the increasing demand for more expressive visual features, Vision Transformers (ViT) [7] have been widely adopted in computer vision tasks [12]. Their success is largely attributed to the self-attention mechanism, which enables each token to aggregate information from all spatial positions, giving ViT a global theoretical receptive field and allowing it to capture globally aggregated contextual semantics [25], as illustrated in Figure 1(b), even though

the effective contributions from different positions may not be equally strong. Owing to these advantages, ViT-based models have achieved state-of-the-art performance in a wide range of visual tasks, including image classification [38], segmentation [28] and image retrieval [3]. However, in hashing-based retrieval, image descriptors are heavily compressed and then quantized, which can lead to the loss of crucial discriminative information, especially under low-bit encoding, when the descriptors mainly emphasize global semantics. To address this issue, we introduce local prompts to highlight discriminative information in image descriptors. These prompts guide the backbone to learn importance-aware image descriptors that place greater emphasis on discriminative information, alleviating information loss during hashing.

Meanwhile, recent advances in state space models (SSMs), exemplified by Mamba [10], have opened new possibilities for balancing computational efficiency and model performance. Building on this progress, VMamba [24] introduces a 2D selective scan (SS2D) into SSMs, enabling spatial traversal and selective aggregation, which makes the architecture better suited for vision tasks. As a result, visual SSMs can achieve broad spatial coverage in a traversal sense, while their effective influence remains locality-biased and path-dependent on scan-path token interactions. This characteristic is partly attributed to HiPPO-initialized state-space dynamics, which induce decaying convolutional kernels, and is further reinforced by the linear scanning and feature selection mechanisms that prioritize spatially neighboring tokens [11]. Such a property makes Mamba a promising backbone for deep hashing, since local discriminative information can help preserve feature discriminability after compression and quantization. Along this line, MambaHash [14] adopts Mamba as the feature extractor to efficiently encode local visual information. However, in practical scenarios, we observe that Mamba may be less robust when the input contains complex visual distractions, such as the background ball shown in Figure 1(c). Motivated by this observation, we integrate the complementary strengths of ViT and Mamba and propose a prompt-based feature interaction paradigm for deep hashing. In this paradigm, Mamba provides locally discriminative prompts to guide ViT in extracting class-level global semantics, enabling compact descriptor representations while improving robustness to visual interference.

Based on the above discussion, we design a hybrid hashing framework driven by dynamic prompting, termed Mamba dynamically prompted Vision Transformer hashing (MpVH). Unlike existing ViT-based [3] or Mamba-based [14] hashing methods, MpVH enables efficient feature interaction between Mamba and ViT through prompt learning. It not only captures global structural semantics via ViT, but also selectively enhances locally discriminative information through Mamba prompter, improving both the global consistency and local discriminability of image descriptors. This is particularly important for reducing the loss of discriminative information during descriptor compression and quantization in hashing. The toy example in Figure 1(c) illustrates the motivation behind our design. Moreover, to bridge the semantic discrepancy caused by the distinct feature extraction behaviors of ViT and Mamba, we introduce a dynamic mixture-of-experts adapter (MoEA). MoEA employs an MoE routing strategy [9] to adaptively model long-range dependencies among prompts and construct informative prompt features, while the adapter aligns the semantic representations produced by

the two architectures, enabling more effective and stable feature interaction. Our main contributions are summarized as follows.

- We leverage the different sensitivities of ViT and Mamba to visual information and propose MpVH, a Mamba dynamically prompted Vision Transformer hashing framework that better preserves discriminative information under compact hash codes.
- We propose a dynamic MoEA that aggregates historical prompts through MoE routing strategy and injects them into the ViT backbone via an adapter, enabling long-range prompt dependency modeling and more effective cross-architecture feature interaction.
- We conduct extensive experiments on three benchmark datasets. The results show that MpVH achieves strong retrieval performance compared with existing deep hashing methods while maintaining competitive model size and encoding efficiency.

2 Related Work

2.1 Vision Transformer and Mamba

Vaswani et al. [31] introduced the Transformer architecture with a self-attention mechanism, which was initially developed for natural language processing. Building on this foundation, Dosovitskiy et al. [7] extended the Transformer architecture to visual tasks and proposed Vision Transformer. ViT partitions an input image into 16×16 patches and processes them as a token sequence, enabling global structural semantics to be captured through self-attention across all patches. This design has led to strong performance in a wide range of computer vision tasks. Since then, ViT has been widely adopted for high-quality global visual feature extraction in various vision applications [13].

Mamba, like the Transformer, was also initially developed for natural language processing [32]. Gu et al. [10] introduced Mamba with a selective mechanism and hardware-aware design, demonstrating strong effectiveness in long-sequence modeling. To extend Mamba to visual tasks, He et al. [40] proposed Vision Mamba, which adopts a bidirectional state space model for visual representation learning from images. Building on this idea, Liu et al. [24] further developed VMamba by introducing an SS2D module that performs four-directional scanning over images. More generally, Mamba's HiPPO-initialized state-space dynamics and spatial scanning mechanism enable it to capture locality-biased discriminative details. Under multi-path scanning, it can also attain broad spatial coverage, although its effective influence remains path-dependent [11].

ViT has demonstrated strong capability in capturing globally aggregated contextual semantics through token-wise global interactions [25], whereas Mamba is more effective at extracting locality-biased discriminative information due to its path-dependent effective influence [24]. However, existing works typically use ViT and Mamba in isolation and seldom integrate them within a unified framework to explicitly leverage their different sensitivities to visual information. This limitation is particularly pronounced in hashing-based image retrieval, where compact hash codes require retaining locally discriminative information without sacrificing global structural semantics.

2.2 Prompt Learning

Prompt learning [34] introduces task-relevant prompts to steer pretrained models toward downstream tasks, improving feature representation and adaptability with only minimal parameter updates. Initially developed in natural language processing, prompt learning has since been successfully extended to various computer vision tasks, such as image classification [39] and segmentation [17]. For example, Cao et al. [1] proposed a multimodal visual prompt tracking framework based on a universal bidirectional adapter to overcome the limitations of single-sensor object tracking. Jia et al. [16] introduced visual prompt tuning by inserting a small number of learnable prompt tokens into ViT, significantly improving the quality of the extracted image features.

In this work, we extend prompt learning to supervised deep hashing, where prompt tokens serve as an effective bridge for feature interaction between heterogeneous architectures. Specifically, they enable the local information captured by Mamba to guide a ViT-based hashing backbone in a single-modality setting.

2.3 Hash Retrieval

With the rapid development of deep hashing, deep learning-based methods have gradually become the mainstream and are generally categorized into unsupervised [30] and supervised hashing [37]. In the unsupervised setting, Li et al. [22] proposed the Bi-Half network, which employs a parameter-free layer to directly map continuous features into binary codes with maximum entropy under a “half-half distribution” constraint. While these methods can effectively learn hash functions without semantic labels or side information, they may encounter the semantic gap issue, potentially degrading the quality of the generated hash codes. By contrast, supervised deep hashing learns hash functions from labeled data by minimizing the Hamming distance between semantically similar samples while maximizing it for dissimilar ones. Wei et al. [35] proposed attribute-aware deep hashing, which establishes explicit correspondences between hash codes and visual attributes to improve fine-grained image retrieval. Li et al. [19] introduced deep supervised discrete hashing, which preserves semantic similarity while enabling category prediction, alleviating the conflict between continuous relaxation and discrete optimization. Benefiting from the strong global representation capability of ViT, Li et al. [21] proposed a multi-scale ViT-based hashing method, which employs a dual-branch architecture to extract and fuse multi-scale features, enhancing the representational capacity of image descriptors.

Previous deep hashing methods [8] have primarily focused on optimizing similarity relationships in the Hamming space, while paying limited attention to the representational quality of image descriptors. As a result, when descriptors are compressed and quantized into compact binary codes, discriminative information may be lost, especially under short code lengths [6]. In contrast, MpVH leverages prompt learning to enable effective feature interaction between components with complementary sensitivities to visual information, allowing the learned descriptors to capture global structural semantics while selectively attending to locally discriminative information. This is important for learning importance-aware image descriptors and alleviating the loss of discriminative information during the hashing process. Moreover, unlike existing

ViT-based or Mamba-based hashing methods, MpVH unifies the two architectures within a single prompting framework to exploit their complementary strengths in visual modeling. Different from prior hybrid hashing frameworks [4], we further introduce MoEA to coordinate cross-architecture feature interaction and adaptively select relevant local prompts.

3 Methodology

3.1 Overall Framework and Problem Definition

MpVH is designed to compress the visual representations of image descriptors through deep interaction between locally discriminative prompts and global structural semantics, alleviating the loss of critical visual information during hashing. As shown in Figure 2, MpVH consists of five main components: an image embedding module, a Mamba prompter, a MoEA, a ViT backbone and a hashing module.

Specifically, MpVH first uses a Mamba prompter to extract locally discriminative prompts Z^{Mamba} from the input image. These prompts are then processed by a MoEA, which models long-range dependencies and bridges the semantic gap between features extracted by different architectures. Finally, the adapted prompts Z^{Prompt} guide the ViT backbone to produce image descriptors with global consistency and local discriminability, alleviating information loss introduced by compression and quantization in hash encoding. In this work, the training set is represented as $T = \{(X_{train}, Y_{train})\}_{i=1}^N$, where X denotes the sample set and Y is the label set. $S \in \{0, 1\}^{N \times N}$ represents the similarity matrix and $S_{ij} \in S$ indicates the pairwise similarity between samples. In addition, MpVH is not restricted to a particular instantiation of Mamba or ViT, but rather provides a reusable design paradigm for leveraging heterogeneous architectures in deep hashing.

3.2 Image Embedding

For a given input image, we follow the ViT strategy [7] to divide it into a series of 2D patches. These patches are then flattened and projected into a sequence of 1D tokens through the embedding module, with positional encoding added. This process can be defined as:

$$z_0^{Mamba} = \text{Embedding}(x) = [x_p^1 E, \dots, x_p^M E] + E_{Pos.}, \quad (1)$$

where z_0^{Mamba} represents the flattened image embedding features with positional encoding, which are used as the input to the Mamba prompter. x_p denotes the split image patches, M is the number of patches. E is the projection matrix, and $E_{Pos.}$ represents the positional encoding.

For the ViT backbone used in image classification tasks, a classification token X_{class} is appended to the image embedding features:

$$z_0^{Trans} = [x_{class}; z_0^{Mamba}], \quad (2)$$

where X_{class} denotes a learnable classification token, which is ultimately used as the sole image descriptor for hash code generation. z_0^{Trans} represents the input to the ViT backbone.

3.3 Mamba Prompter

In the Mamba prompter, we leverage the stronger locally information sensitivity of SSMs [10]. Leveraging the globally aggregated

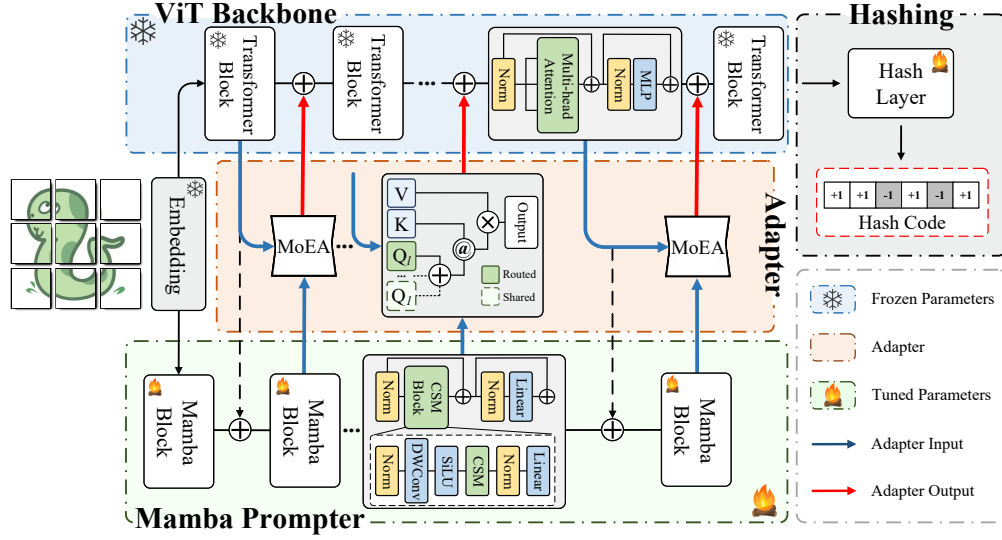


Figure 2: MpVH architecture.

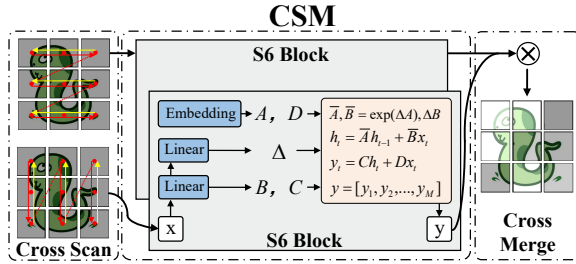


Figure 3: Cross scan model architecture.

contextual semantics captured by ViT, the prompter selectively extracts locally discriminative information as prompts to guide the feature learning of the ViT backbone.

Specifically, the Mamba prompter consists of L Mamba blocks. The first block takes the z_0^{Mamba} as input, while each of the remaining $(L - 1)$ blocks receives a residual combination of the previous prompter output and the ViT output z_{l-1}^{Mamba} and the ViT output z_{l-1}^{Trans} . This process can be expressed as:

$$z_l^{Mamba} = \Phi_l^{Mamba} \left(z_{l-1}^{Trans} + z_{l-1}^{Mamba} \right), \quad l > 1, \quad (3)$$

where z_l^{Mamba} indicates the Mamba prompt at l -th layer, and Φ_l^{Mamba} represents the structure of the Mamba prompter as illustrated in Figure 2. This fusion process is performed only on the patch tokens.

Although the existing SS2D strategy [24] is capable of modeling local dependencies, its reliance on a single traversal path may reduce robustness to noise. To enhance the robustness of prompt extraction, we propose a cross scan model (CSM), which revisits local regions through bidirectional cyclic scanning to enlarge spatial coverage while preserving locality-biased effective influence, as shown in Figure 3. Specifically, CSM performs bidirectional cyclic scanning over the input features along both spatial dimensions, allowing it to capture more locally discriminative information.

The S6 block performs iterative feature selection, while the cross merge integrates the selected features to enhance the representation of locally discriminative information. The selection process for the S6 block is formulated as follows:

$$z_{l,m}^{Mamba} = Ch_t + D \left(z_{l-1,m}^{Trans} + z_{l-1,m}^{Mamba} \right), \quad (4)$$

$$h_t = \bar{A}h_{t-1} + \bar{B} \left(z_{l-1,m}^{Trans} + z_{l-1,m}^{Mamba} \right), \quad (5)$$

where $z_{l,m}^{Mamba}$ refers to the m -th token after feature selection, and M denotes the total number of tokens. h_t denotes the intermediate hidden state function, which stores the contextual information among tokens. $\bar{A} = \exp(\Delta A)$, $\bar{B} = \Delta B$, C and D are all projection matrices.

Formally, given a 2D feature map, we linearize it into multiple 1D scan sequences along different spatial directions. Each sequence is independently processed by Mamba to model scan-path-specific dependencies. The resulting outputs are subsequently reshaped back to the 2D layout and combined via a cross-merge operation, which aggregates complementary information across traversal paths. Through multi-directional revisiting of local regions, the proposed CSM captures more robust and discriminative local cues than the original VMamba, while maintaining linear-time complexity.

3.4 Dynamic Mixture-of-Experts Adapter

The image features extracted from two different networks exhibit semantic gaps, and the direct application of prompts may lead to training instability. To bridge this gap and adaptively model long-range dependencies among prompts, we propose a dynamic MoEA, which comprises a MoE module and an adapter, as illustrated in Figure 4.

Mixture-of-experts module: The MoE dynamically selects shared and routed experts and assigns adaptive fusion weights to the prompts at different layers. Different from the conventional use of MoE [15] for sparsifying model computation during training, we

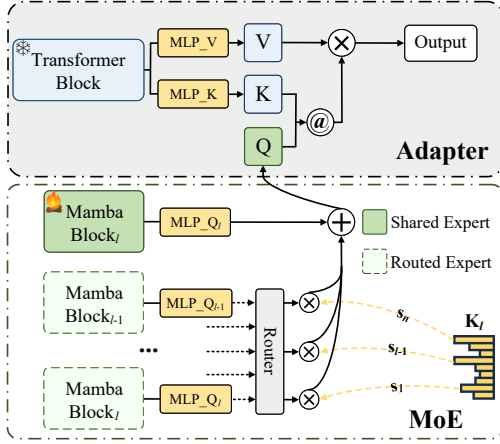


Figure 4: Dynamic mixture-of-experts adapter architecture.

adopt MoE as a prompt routing mechanism to model dependencies among historical prompts and to suppress redundant prompts.

First, the Mamba prompt z_l^{Mamba} is selected as the shared expert and is employed to participate in the adapted prompt computation at the l -th layer. The mapping process for each expert prompt is defined as:

$$Q_l = \varphi_Q \left(z_l^{Mamba} \right), \quad (6)$$

where Q_l represents the query obtained through the shared expert mapping, which helps reduce semantic gaps. φ_Q is a mapping network with a multi-layer perceptron structure, implemented using a 1×1 convolution.

To dynamically select historical prompts that are most relevant to the current ViT tokens, we adopt an MoE routing strategy. Specifically, we treat the $(l-1)$ previously generated Mamba prompts as routed experts Q_n and activate those whose similarity scores exceed ε , resulting in an adaptive number of routed experts K_l at layer l . This routing strategy injects informative historical context while maintaining sparse computation. We define the routed-expert selection as:

$$\dot{s}_n = \frac{\sigma(Q_n^T V_{l-1})}{\sum_{n=0}^{l-1} \sigma(Q_n^T V_{l-1})}, \quad (7)$$

$$s_n = \begin{cases} \dot{s}_n, & \dot{s}_n \geq \varepsilon \\ 0, & \dot{s}_n < \varepsilon, \end{cases} \quad (8)$$

where s_n denotes the weighting coefficient for the n -th routed expert, $\dot{s}_n$ is the normalized feature similarity score, ε is the similarity threshold that controls which experts are activated, and $\sigma(\cdot)$ is a Sigmoid function. A routed expert is activated and assigned a weight s_n only when the gating value satisfies $\dot{s}_n \geq \varepsilon$; otherwise, its weight is set to zero.

Finally, we use s_n as weighting coefficients to fuse the shared and routed experts. This produces the query $Q^{(l)}$ used in the adapted prompts computation at the l -th layer:

$$Q^{(l)} = Q_l + \sum_{n=0}^{l-1} s_n Q_n, \quad (9)$$

where $Q^{(l)}$ indicates the low-redundancy final query with long-range dependencies. At each layer, the number of shared experts used for prompting is fixed to one, while the number of routed experts is adaptive and determined by the similarity threshold ε .

Adapter: The adapter employs an interactive fusion mechanism to align the final query $Q^{(l)}$ with the output of the ViT, ultimately generating the adapted prompt z_l^{Prompt} that meets the input requirements of the ViT backbone.

Specifically, for the ViT output at the $(l-1)$ -th layer, we project the features into key (K) and value (V). Subsequently, an interactive fusion mechanism is applied to perform attention computation over Q , K , and V :

$$z_l^{Prompt} = \text{Softmax} \left(\frac{Q^{(l)} K_{l-1}^T}{\sqrt{d}} \right) V_{l-1}, \quad (10)$$

where z_l^{Prompt} denotes the adapted prompt generated at the l -th layer. $K_{l-1} = \varphi_K(z_{l-1}^{Trans})$, $V_{l-1} = \varphi_V(z_{l-1}^{Trans})$, and d is the scaling factor. φ_K and φ_V adopt an MLP structure and are implemented using 1×1 convolution for feature mapping.

Through the coordinated interaction of these two components, MoEA not only mitigates the semantic discrepancy between heterogeneous architectures, but also models dependencies among historical prompts to enhance prompt quality, facilitating more effective feature interaction for learning compact yet expressive image descriptors.

3.5 Vision Transformer Backbone

After obtaining the adapted prompt z_l^{Prompt} , we fuse it with the previous layer ViT tokens z_{l-1}^{Trans} , and feed the combined tokens into the l -th Transformer block.

$$z_l^{Trans} = \Phi_l^{Trans} \left(z_{l-1}^{Trans} + z_l^{Prompt} \right), \quad (11)$$

where z_l^{Trans} indicates the output feature of the l -th Transformer block, and Φ_l^{Trans} refers to the Transformer block as illustrated in Figure 2.

The ViT backbone consists of L Transformer blocks. We extract the classification token x_{class} from the output z_L^{Trans} of the final block as the image descriptor. The final image descriptor balances the global structural semantics provided by the ViT backbone and the locality-biased discriminative prompts injected by the Mamba prompter, thereby helping preserve more discriminative information during the hashing process.

3.6 Hashing Module and Training

Finally, we feed the image descriptor x_{class} into the hashing module to generate a compact binary hash code b_i :

$$b_i = \text{sign} \left(f_{hash} (x_{class}; \theta_{hash}) \right), \quad (12)$$

where x_{class} denotes the input image descriptor, $f_{hash}(\cdot)$ represents the hashing network with parameters θ_{hash} , $\text{sign}(\cdot)$ is the sign function, and b_i is the binary hash code.

However, if the symbolic function $\text{sign}(\cdot)$ is used directly to convert the image descriptor to a binary hash code, we are unable to compute gradients for optimization. Therefore, we approximate the symbolic function using the hyperbolic tangent function $\tanh(\cdot)$, as shown in Eq. (13). In addition, we introduce a quantization loss $\mathcal{L}_Q$

(see Eq. (14)) to minimize the discrepancy between the generated real-valued hash code and the desirable hash code.

$$h_i = \tanh(f_{\text{hash}}(x_{\text{class}}; \theta_{\text{hash}})), \quad (13)$$

$$\mathcal{L}_Q = \sum_{i=1}^N \|h_i - b_i\|_2^2, \quad (14)$$

where h_i denotes the final generated hash code, and $\tanh(\cdot)$ represents the hyperbolic tangent function. $\mathcal{L}_Q$ refers to the quantization loss, and b_i is the stop-gradient binary target.

To further ensure the quality of the generated hash codes, we design a hash loss function $\mathcal{L}_H$ to preserve the semantic similarity between different samples in the Hamming space. The definition of $\mathcal{L}_H$ is provided as:

$$\mathcal{L}_H = \sum_{i=1}^N \sum_{j=1}^N \left(h_i^T h_j - k S_{ij} \right)^2, \quad (15)$$

where $h_i^T h_j$ is the inner product of two group hash codes, $S_{ij} \in \{0, 1\}$ indicates the pairwise similarity between samples, and k is a scaling factor. The total loss is defined as:

$$\mathcal{L} = \lambda_1 \mathcal{L}_H + \lambda_2 \mathcal{L}_Q, \quad (16)$$

where $\mathcal{L}$ denotes the total loss, and λ_1 and λ_2 are the weighting coefficients. **Appendix A** provides the pseudocode.

4 Experiment

4.1 Datasets and Experiment Settings

We conduct extensive experiments on three benchmark datasets to evaluate the retrieval performance of MpVH, including CIFAR-10 [18], ImageNet [29], and NUS-WIDE [5]. The detailed descriptions of the datasets are provided in **Appendix B**.

Our MpVH model is implemented using the PyTorch framework, and all experiments are conducted on a workstation equipped with an Intel i9-12900K CPU and two GeForce RTX 3090-OX24G GPUs. During training, we adopt ViT-B as the backbone and do not completely freeze the image embedding module and ViT backbone; instead, we apply a small learning rate for fine-tuning, with the initial learning rate set to 3.0×10^{-6} and a weight decay of 1.0×10^{-9} . For the Mamba prompt, the MoEA, and the hashing module, we adopt standard training settings, using an initial learning rate of 3.0×10^{-3} and a weight decay of 1.0×10^{-6} . Across all datasets, we adopt SGD as the optimizer with a batch size of 16 and train the model for 300 epochs. Based on empirical observations, we set the hyperparameters in MpVH to $\varepsilon = 0.6$ and $\lambda_1 = \lambda_2 = 1$.

4.2 Main Experiment

In the comparative experiments, we adopt three widely used metrics to evaluate the retrieval performance of MpVH: mean average precision (mAP), precision-recall (PR) curves, and precision at the top- N returned samples (P@N). Table 1 reports the mean mAP results of all compared methods over five runs under different random seeds and code lengths. MpVH achieves competitive performance against existing methods based on ResNet, ViT, or Mamba backbones, with more advantages on the more challenging datasets.

This improvement can be attributed to the prompt learning strategy of MpVH, which integrates the global structural semantics captured by ViT with the locally discriminative prompts generated by Mamba, producing more importance-aware image descriptors and more discriminative hash codes.

Figure 5 presents the PR and P@N curves of various deep hashing methods on the CIFAR-10, ImageNet, and NUS-WIDE datasets under 16-bit and 64-bit code lengths. It can be observed that MpVH remains competitive with other hashing methods by achieving a larger area under the PR curve and maintaining higher precision across the P@N curve. These results further demonstrate the effectiveness of MpVH for deep hashing-based image retrieval.

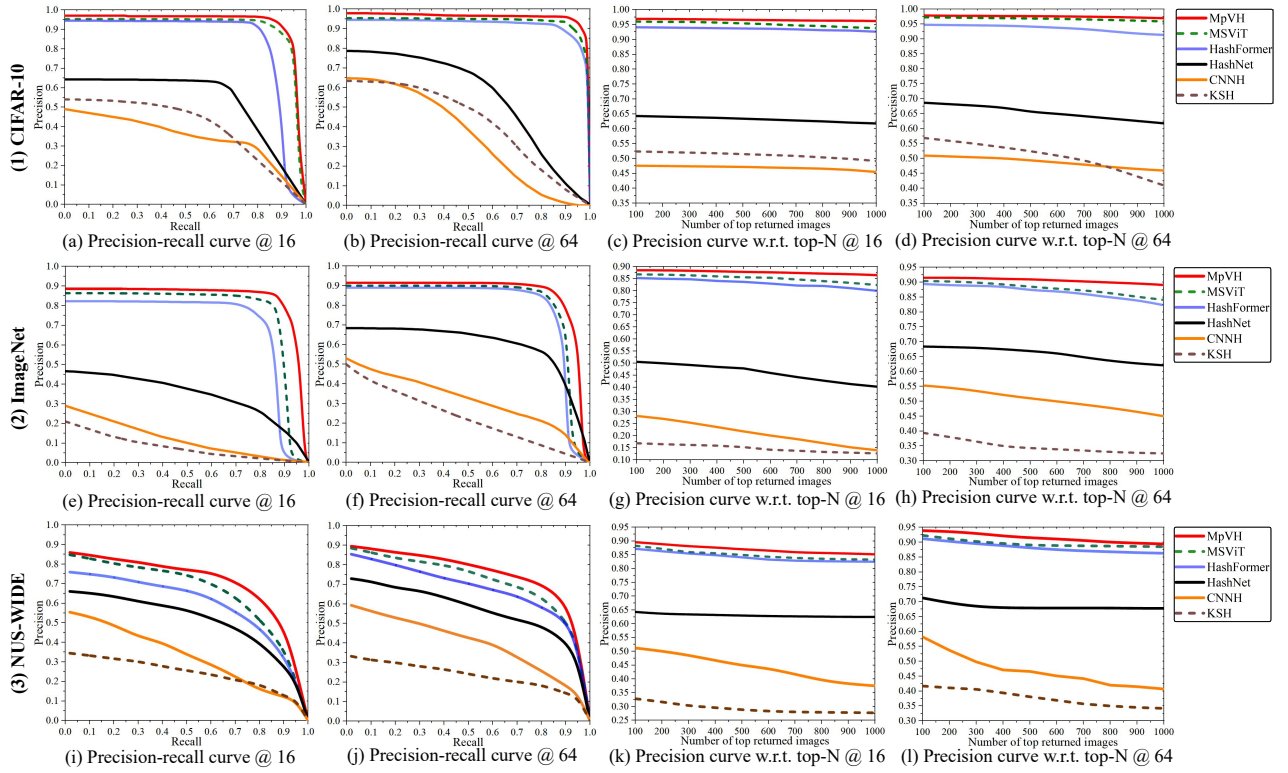
4.3 Ablation Experiment

MoEA is designed to facilitate stable and effective feature interaction between the Mamba prompt and the ViT backbone. To assess the contribution of MoEA and its components, we conduct ablation studies by removing the entire MoEA module, the MoE block, or the adapter, as shown in Figure 6. When MoEA or the adapter is removed, the adapter is replaced with a simple MLP to avoid training instability caused by the output discrepancy between ViT and Mamba. Figure 6 reveals three key findings. First, removing the MoE block (w/o MoE) causes a noticeable drop in retrieval performance, showing that MoE improves prompt quality by modeling dependencies among historical prompts while suppressing redundant information through sparse expert selection. We further study the similarity threshold ε for prompt sparsity in Figure 7. When $\varepsilon = 0$, all historical prompts are softly aggregated, which introduces excessive and potentially irrelevant information; when $\varepsilon = 1$, only the current-layer prompt is retained, limiting cross-layer dependency modeling. Therefore, we set $\varepsilon = 0.6$ to balance sparsity and dependency modeling. Second, replacing the adapter with a simple MLP (w/o adapter) further reduces retrieval accuracy, indicating that the adapter is crucial for bridging the semantic gap between Mamba and ViT features. Finally, removing the entire MoEA module (w/o MoEA) yields the worst performance, suggesting that simple MLP-based interaction is insufficient for coordinating the two heterogeneous architectures. These results confirm the effectiveness of MoEA and the proposed threshold-based routing strategy.

We design MpVH based on the observation that different network architectures exhibit distinct sensitivities to visual information. To justify the choice of using Mamba as the prompt and ViT as the backbone in MpVH, we conduct extensive ablation studies on the ImageNet dataset. Specifically, we construct different model variants by replacing the backbone and prompter to evaluate the effectiveness of different backbone–prompter combinations, as reported in Table 2. The results show that the combination of ViT as the backbone and Mamba as the prompter achieves the highest retrieval accuracy. This observation highlights the strong complementarity between the globally aggregated semantics captured by ViT and the locally discriminative prompts generated by Mamba. Such synergy enables MpVH to effectively compress visual representations, guide attention toward highly discriminative regions, and reduce information loss during hash encoding. In contrast, although some other combinations also achieve reasonably good retrieval performance, they remain inferior to MpVH. This result

Table 1: mAP comparisons of different methods on three benchmark datasets with different bits.

Method	CIFAR-10(mAP@5000)				ImageNet(mAP@1000)				NUS-WIDE(mAP@1000)			
	16 bits	32 bits	48 bits	64 bits	16 bits	32 bits	48 bits	64 bits	16 bits	32 bits	48 bits	64 bits
KSH [23]	0.5239	0.5583	0.5668	0.5688	0.1599	0.2976	0.3422	0.3943	0.3561	0.3327	0.3124	0.3368
CNNH [36]	0.4756	0.4723	0.4886	0.5013	0.2812	0.4498	0.5245	0.5538	0.5696	0.5827	0.5926	0.5996
HashNet [2]	0.6431	0.6672	0.6751	0.6873	0.5059	0.6306	0.6633	0.6835	0.6623	0.6988	0.7114	0.7163
HashFormer(Res) [20]	0.8542	0.8661	0.8719	0.8745	0.7215	0.8422	0.8575	0.8690	0.6982	0.7124	0.7230	0.7248
HashFormer(ViT) [20]	0.9121	0.9167	0.9211	0.9263	0.7791	0.8962	0.9007	0.9010	0.7317	0.7418	0.7592	0.7597
TransHash [3]	0.9075	0.9108	0.9141	0.9166	0.7852	0.8733	0.8932	0.8921	0.7263	0.7339	0.7532	0.7448
SWTH [26]	0.9539	0.9616	0.9692	0.9734	0.7837	0.8779	0.8870	0.9041	0.7462	0.7531	0.7708	0.7829
MSViT [21]	0.9473	0.9565	0.9610	0.9692	0.8696	0.8825	0.8934	0.9036	<u>0.8523</u>	<u>0.8711</u>	<u>0.8792</u>	<u>0.8902</u>
MambaHash [14]	0.9540	0.9594	0.9677	0.9719	0.8133	<u>0.8974</u>	<u>0.9081</u>	<u>0.9133</u>	0.8375	0.8566	0.8643	0.8680
ICVT [4]	0.9696	0.9728	0.9732	0.9745	0.8726	0.8969	0.9002	0.9041	0.8411	0.8629	0.8689	0.8733
MpVH (Ours)	<u>0.9689</u>	0.9741	0.9746	0.9791	0.8851	0.8998	0.9088	0.9155	0.8624	0.8779	0.8817	0.8963

**Figure 5: The experimental results of different methods on three benchmark datasets under PR and P@N evaluation metrics.****Table 2: Backbone-prompter combination ablation.**

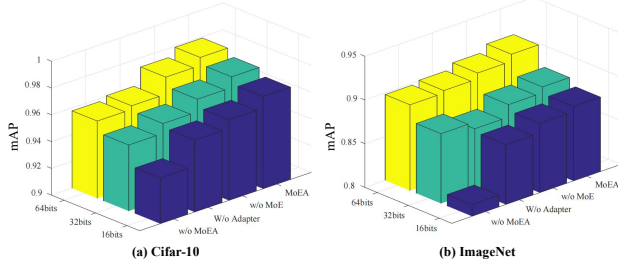
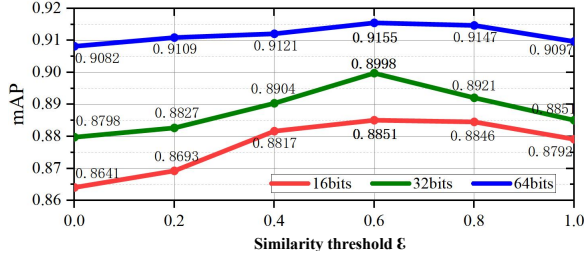
Backbone		Prompter			ImageNet(mAP)		
ViT	Mamba	ResNet	ViT	Mamba	16bits	32bits	64bits
✓		✓			0.8421	0.8685	0.8924
✓			✓		0.7677	0.8229	0.8533
✓				✓	0.8851	0.8998	0.9155
	✓	✓			0.6885	0.7533	0.8229
	✓		✓		0.8577	0.8709	0.8931
	✓			✓	0.7569	0.8307	0.8492

further supports the effectiveness of the configuration with Mamba as the prompter and ViT as the backbone, which is also corroborated by the subsequent visualization results.

The scanning pattern plays a crucial role in enabling Mamba to perform effectively on visual tasks. To evaluate the impact of this design choice in the Mamba prompter, we compare several representative scanning strategies, including unidirectional, bidirectional, oblique, SS2D, and our proposed CSM, as summarized in Table 3. The results show that CSM achieves the best retrieval performance across different datasets and code lengths. We attribute this improvement primarily to the bidirectional cyclic scanning

Table 3: Performance comparison of different scanning patterns in the Mamba prompter.

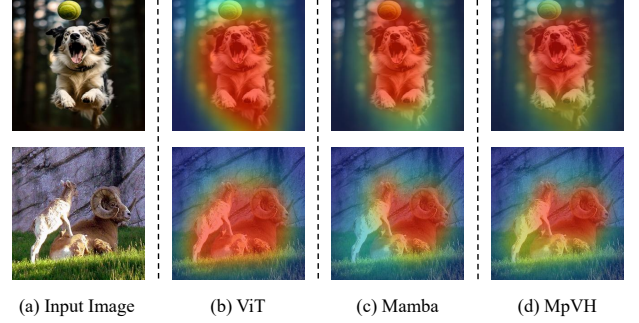
Scanning Patterns	CIFAR-10 (mAP@5000)			ImageNet (mAP@1000)			NUS-WIDE (mAP@1000)		
	16bits	32bits	64bits	16bits	32bits	64bits	16bits	32bits	64bits
Unidirectional scanning	0.9403	0.9481	0.9601	0.8596	0.8633	0.8847	0.8212	0.8429	0.8504
Bidirectional scanning	0.9521	0.9595	0.9708	0.8776	0.8859	0.9011	0.8513	0.8698	0.8882
Oblique scanning	0.9638	0.9722	0.9761	0.8827	0.8904	0.9081	0.8592	0.8743	0.8952
SS2D	0.9631	0.9729	0.9774	0.8824	0.8898	0.9127	0.8605	0.8751	0.8959
CSM (Ours)	0.9689	0.9741	0.9791	0.8851	0.8998	0.9155	0.8624	0.8779	0.8963

**Figure 6: Ablation study on MoEA components.****Figure 7: Analysis of the similarity threshold ϵ hyperparameter.**

mechanism in CSM, which allows local features to be revisited through multiple spatial traversal paths. Compared with SS2D, this design enlarges the spatial coverage of the Mamba prompter by revisiting local regions through multiple traversal paths, allowing it to capture more diverse local information while preserving locality-biased effective influence. In contrast, unidirectional, bidirectional, and oblique scanning patterns impose stronger directional biases, which may limit the diversity of local context aggregation.

4.4 Visualization Result

We visualize the response maps of different networks to better understand their distinct feature sensitivities. According to the qualitative analysis in **Appendix E**, ViT captures globally aggregated contextual semantics through token-wise global interactions, whereas the Mamba prompter, even under cross-scan traversal, retains locality-biased and path-dependent effective influence. This distinction is clearly reflected in Figure 8. As shown in Figure 8(b), ViT primarily emphasizes class-level global structural semantics, which is consistent with its self-attention mechanism, where each

**Figure 8: Visualization of attention maps in three networks.**

token aggregates information from all spatial positions. For example, it can largely ignore the irrelevant balls in the background and focus on the overall semantics of the object. In contrast, the Mamba model in Figure 8(c) places greater emphasis on fine-grained local information, owing to its locality-biased effective influence induced by selective state-space propagation and cross-scan traversal. As a result, Mamba captures more discriminative local information for distinguishing similar instances, although it may be less robust when the background is complex or noisy. As shown in Figure 8(d), MpVH combines the strengths of both architectures by integrating global structural semantics with locally discriminative prompts, improving the discriminative representation of image descriptors.

5 Conclusion

In this study, we proposed MpVH, a prompt learning framework for deep hashing that leverages the differing sensitivities of network architectures to visual information in order to reduce discriminative information loss during hash encoding. Specifically, MpVH uses a Mamba prompter to extract locality-biased discriminative prompts that guide a ViT backbone to learn more compact image descriptors. To enable stable and effective interaction between the two architectures, we further introduced a dynamic MoEA, where the MoE models long-range dependencies among prompts while suppressing redundancy, and the adapter bridges the semantic gap between heterogeneous features. By enhancing the discriminative representation of image descriptors, MpVH alleviates the information loss caused by compression and quantization during hashing. Extensive experiments on benchmark datasets showed that MpVH achieves competitive retrieval performance while maintaining a compact model size. In future work, we will investigate more efficient descriptor representations.

Acknowledgments

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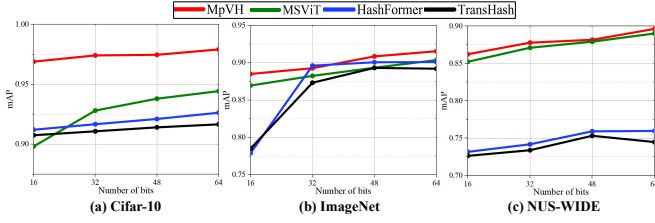


Figure 9: The experimental results of different methods on three benchmark datasets under mAP-bits metrics.

Algorithm 1 The training procedure of MpVH

Input: Dataset $T = \{X^{train}, Y^{train}\}_{i=1}^N$ and similarity matrix S .

Parameter: Hyperparameters L , λ , and θ .

Output: Hash code h_i .

- 1: **Embedding module:** $x \rightarrow z_0^{Mamba}, z_0^{Trans}$, according to Eqs. (1) and (2);
 - 2: **for** $l \in L$ **do**
 - 3: **Mamba prompter:** $z_{l-1}^{Mamba}, z_{l-1}^{Trans} \rightarrow z_l^{Mamba}$, according to Eqs. (3) to (5);
 - 4: **MoEA:** $z_{l-1}^{Trans}, z_l^{Mamba} \rightarrow z_l^{Prompt}$, according to Eqs. (6) to (10);
 - 5: **ViT backbone:** $z_{l-1}^{Trans}, z_l^{Prompt} \rightarrow z_l^{Trans}$, according to Eq. (11);
 - 6: **end for**
 - 7: **Hashing module:** $z_L^{Trans} \rightarrow x_{class} \rightarrow h_i$;
 - 8: Binary hash code: $x_{class} \rightarrow b_i$, according to Eq. (12);
 - 9: Hash code: $x_{class} \rightarrow h_i$, according to Eq. (13);
 - 10: **Return** h_i .
-

A Training Procedure

In the training phase, the model is trained in an end-to-end manner. In the testing phase, MpVH is used as the hash function to map input images into binary hash codes. The overall training procedure of MpVH consists of five steps.

First, through the image embedding module, each input image $X \in T$ is divided into a set of patches. Each patch is then embedded with content and positional encodings to generate a token sequence suitable for both ViT and Mamba. Second, in the Mamba prompter, the outputs of the previous prompter layer and the current ViT layer are fused through weighted summation. The fused features are then fed into the CSM within the prompter for feature selection, producing the Mamba prompts Z^{Mamba} . Third, in the dynamic MoEA, both the Mamba prompts and the ViT outputs are fed into the MoEA to generate adapted prompts Z^{Prompt} . Specifically, the MoEA adaptively establishes long-range dependencies among the Mamba prompts through the mixture-of-experts module and produces normalized adapted prompts through the adapter. Fourth, the ViT backbone uses the adapted prompts to guide feature extraction in each Transformer block, yielding progressively more discriminative image features Z^{Trans} . These features aggregate global information through self-attention while emphasizing locally discriminative cues via the selection mechanism, resulting in compact and expressive image descriptors. Fifth, the hashing

module quantizes the final image descriptor to generate hash codes h , which preserve strong discriminative capability in the Hamming space for efficient retrieval. The overall training process of MpVH is summarized in Algorithm 1.

B Datasets and Experiment Settings

CIFAR-10: The dataset consists of 10 categories, each containing 6,000 RGB images of size 32×32 . Following the experimental protocol in [21], we randomly select 1,000 images in total (100 per category) as query samples, randomly sample 5,000 images from the remaining ones as the training set, and use the rest as the retrieval database.

ImageNet: This dataset serves as a benchmark for the ILSVRC 2015 large-scale visual recognition challenge, comprising 1,000 categories with approximately 1.2 million training images and 50,000 validation images. Following the protocol in [2], we construct the training/retrieval set by randomly selecting 100 categories and sampling 130 images per category. In addition, 5,000 images from the original validation set are used as query samples.

NUS-WIDE: This dataset consists of 269,648 images, each annotated with one or more of 81 ground-truth categories. Following the experimental protocol in [21], we randomly select 5,000 images as query samples, while the remaining images constitute the retrieval database. Additionally, 10,000 images are randomly sampled from the database to form the training set.

For a fair comparison, all ViT-based methods are initialized from the same ImageNet-pretrained ViT backbone. Following the protocols in [2], we report mAP@1000 for ImageNet and NUS-WIDE, and mAP@5000 for CIFAR-10. Moreover, for the NUS-WIDE dataset, we follow the standard evaluation protocol described in [2], where two images are considered similar (i.e., $s_{ij} = 1$) if they share at least one label, and dissimilar (i.e., $s_{ij} = 0$) otherwise.

C Supplementary Experiments

C.1 Hash Code Lengths

Figure 9 illustrates the retrieval performance of MpVH under different hash code lengths on the CIFAR-10, ImageNet, and NUS-WIDE datasets. Compared with other ViT-based hashing methods, MpVH delivers superior and stable retrieval accuracy across all code lengths, particularly under shorter codes. This advantage mainly arises from the selective representation capability of the image descriptors learned by MpVH, which enables the model to focus more effectively on discriminative image regions and thus generate more representative hash codes even under limited code lengths.

C.2 Model Complexity

We compare the retrieval performance, model size, and computational complexity of MpVH with other mainstream deep hashing methods, as reported in Table 4. Specifically, we report the retrieval accuracy (mAP), number of parameters, FLOPs, and hash code generation speed on the ImageNet dataset under the same hash code length. Where, “Extraction/1pic” denotes the speed of extracting image descriptors for a single image during inference. As shown in Table 4, MpVH achieves significantly better retrieval performance than other ViT-based models. Despite incorporating both the Mamba prompter and the MoEA, it still maintains lower

Table 4: Model complexity comparison.

Method	ImageNet(mAP@1000)			Params.	FLOPs	Extraction /1pic
	16bits	32bits	64bits			
HashNet	0.5059	0.6306	0.6835	60.8M	0.73G	0.064s
HashFormer	0.7791	0.8962	0.9010	88.9M	98.5G	0.983s
MSViT	0.8696	0.8825	0.9036	92.3M	102.5G	1.109s
MambaHash	0.8133	0.8974	0.9133	83.9M	82.6G	0.723s
ICVT	0.8726	0.8969	0.9041	93.5M	104.1G	1.062s
MpVH	0.8851	0.8998	0.9155	81.6M	79.3G	0.717s

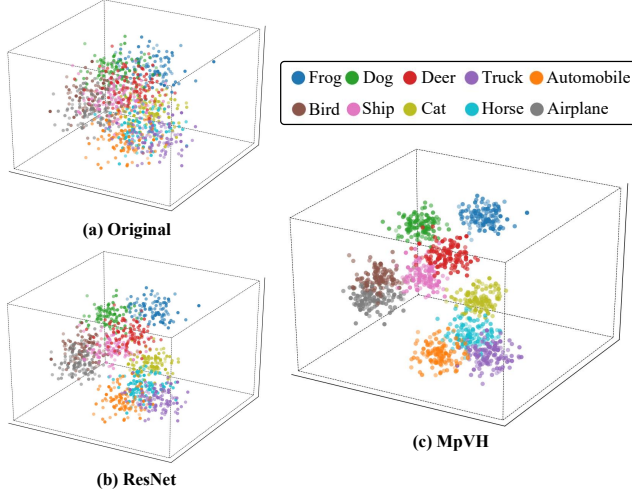


Figure 10: Visualization of class distributions in the image descriptor for different models.

Datasets	Query	Top-8 retrieved images							
CIFAR-10	Ship								
	Bird								
ImageNet	Dog								
	Ostrich								
NUS-WIDE	Flower								
	Bear								

Figure 11: Examples of top-8 retrieved images on three datasets of 64 bits hash codes by MpVH.

Table 5: Sensitivity analysis of λ_1 and λ_2 on ImageNet.

	0.2		0.5		1		2	
	16bits	64bits	16bits	64bits	16bits	64bits	16bits	64bits
λ_1	0.8780	0.9117	0.8805	0.9132	0.8851	0.9155	0.8793	0.9129
λ_2	0.8823	0.9129	0.8818	0.9134	0.8851	0.9155	0.8796	0.9090

computational overhead than most mainstream ViT-based models. These results demonstrate that MpVH achieves a favorable balance between retrieval accuracy and computational efficiency.

C.3 Hyperparameter Sensitivity

To evaluate the robustness of MpVH to different loss-weight settings, we conduct sensitivity experiments for λ_1 and λ_2 on ImageNet using 16bits and 64bits hash codes, with each parameter varied over $\{0.2, 0.5, 1, 2\}$. Specifically, λ_1 controls the preservation of semantic similarity in the Hamming space, while λ_2 regulates the quantization consistency between the generated hash codes and their desirable binary targets. As shown in Table 5, MpVH achieves the best performance when $\lambda_1 = \lambda_2 = 1$. The results show that a small λ_1 weakens the semantic discriminability of the hash codes, whereas a small λ_2 increases the quantization error. In contrast, an excessively large λ_2 overemphasizes binary approximation and suppresses semantic discriminability. Therefore, the adopted setting provides an effective balance between semantic similarity preservation and quantization consistency.

D Supplementary Visualization Results

D.1 Visualization of Class Distributions

To demonstrate the discriminative capability of the image descriptors generated by MpVH, we randomly select 100 samples from each of the 10 classes in the CIFAR-10 dataset for feature visualization, as shown in Figure 10. Figure 10(a) presents the original data distribution without feature extraction, Figure 10(b) shows the image descriptors extracted by the ResNet50 backbone, and Figure 10(c) illustrates the image descriptors generated by MpVH. As can be observed, MpVH produces image descriptors with clearer inter-class boundaries and more compact intra-class clustering than the other methods. This indicates that MpVH effectively leverages the differing sensitivities of ViT and Mamba to visual information, thereby learning more discriminative image descriptors. This not only improves the representational quality of the descriptors but also alleviates information loss during binary hash code generation.

D.2 Visualization of Retrieval Results

Finally, we present the retrieval results of MpVH on CIFAR-10, ImageNet, and NUS-WIDE, as shown in Figure 11. A check mark “✓” denotes a successful retrieval case, while a cross “✗” indicates a failed one. The results show that MpVH achieves strong retrieval

performance and maintains high retrieval accuracy across different datasets. This further demonstrates the robustness and stability of MpVH in practical retrieval scenarios.

E Qualitative Analysis of Complementary Feature Sensitivities

To further justify the design of MpVH, we provide a qualitative theoretical analysis of the complementary feature sensitivities of ViT and Mamba during feature extraction. Our goal is not to establish a strict optimality theorem, but to present a mathematically grounded explanation that is consistent with the empirical observations in the visualization and ablation studies.

Definition 1 (Theoretical and Effective Receptive Fields).

For a deep network f and the i -th output feature $z_i^{(l)}$ at layer l , we define:

- $$\mathcal{R}_{\text{th}}(z_i^{(l)}) = \left\{ j : \frac{\partial z_i^{(l)}}{\partial x_j} \neq 0 \right\}, \quad (17)$$

where x_j denotes the j -th input token (or image patch);

- the ε -effective receptive field as

$$\mathcal{R}_{\text{eff}}^\varepsilon(z_i^{(l)}) = \left\{ j : \left\| \frac{\partial z_i^{(l)}}{\partial x_j} \right\|_2 > \varepsilon \right\}, \quad (18)$$

where $\varepsilon > 0$ is a predefined threshold controlling whether the influence is practically significant.

The theoretical receptive field captures whether a dependency exists, whereas the effective receptive field characterizes whether the dependency is strong enough to be meaningful in practice.

Global Dependency of ViT. Let $\{x_j\}_{j=1}^M$ denote the sequence of image tokens after patch embedding. For simplicity, we consider a single self-attention head:

$$z_i = \sum_{j=1}^M \alpha_{ij} V_j, \quad \alpha_{ij} = \frac{\exp\left(\frac{Q_i K_j^\top}{\sqrt{d}}\right)}{\sum_{k=1}^M \exp\left(\frac{Q_i K_k^\top}{\sqrt{d}}\right)}, \quad (19)$$

where $Q_i = W_Q x_i$, $K_j = W_K x_j$, and $V_j = W_V x_j$.

Under non-degenerate finite logits, the Softmax weights satisfy $\alpha_{ij} > 0$ for all j . Therefore, each output token z_i is computed as a weighted aggregation of all value vectors $\{V_j\}_{j=1}^M$. Moreover, since both V_j and α_{ij} depend on the input tokens, the Jacobian $\partial z_i / \partial x_j$ is non-zero for all j . Hence, for a single Transformer block,

$$\mathcal{R}_{\text{th}}(z_i) = \{1, 2, \dots, M\}. \quad (20)$$

For a ViT composed of L stacked Transformer blocks, the above argument extends inductively, yielding

$$\mathcal{R}_{\text{th}}(\text{ViT}) = \{1, 2, \dots, M\}. \quad (21)$$

Therefore, ViT has a global theoretical receptive field.

This global dependency underlies ViT's ability to capture global structural semantics and preserve class-level contextual consistency. However, a global theoretical receptive field does not imply equal contributions from all spatial positions; rather, it indicates that each position can influence the resulting feature representation.

Locality-Biased Effective Influence of the Mamba Prompter.

Unlike standard time-invariant linear systems, the Mamba prompter is a selective state space model whose parameters vary with the input. A simplified time-varying recurrence can be written as

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = \bar{C}_t h_t, \quad (22)$$

where x_t is the token at scan step t , h_t is the latent state, and $\bar{A}_t, \bar{B}_t, \bar{C}_t$ depend on the current input.

Unrolling the recurrence gives

$$h_t = \left(\prod_{k=1}^t \bar{A}_k \right) h_0 + \sum_{\tau=1}^t \left(\prod_{k=\tau+1}^t \bar{A}_k \right) \bar{B}_\tau x_\tau, \quad (23)$$

where the matrix product is ordered from left to right in increasing time index.

To characterize the effective influence of historical inputs, we introduce the following analysis condition.

Assumption 1 (Uniform Contraction and Bounded Input Gain). Assume there exist constants $0 < \rho < 1$ and $B_{\max} > 0$ such that

$$\|\bar{A}_t\|_2 \leq \rho, \quad \|\bar{B}_t\|_2 \leq B_{\max}, \quad \forall t. \quad (24)$$

This assumption states that the selective dynamics remain uniformly stable during propagation. Under this condition, for any $\tau \leq t$, we have

$$\left\| \frac{\partial h_t}{\partial x_\tau} \right\|_2 \leq \left\| \left(\prod_{k=\tau+1}^t \bar{A}_k \right) \bar{B}_\tau \right\|_2 \leq B_{\max} \rho^{t-\tau}. \quad (25)$$

Therefore, the influence of a past token decays exponentially with its path distance to the current state. Given a threshold $\varepsilon > 0$, define

$$\Delta_\varepsilon = \left\lceil \frac{\log(\varepsilon/B_{\max})}{\log \rho} \right\rceil. \quad (26)$$

Then, whenever $t - \tau > \Delta_\varepsilon$,

$$\left\| \frac{\partial h_t}{\partial x_\tau} \right\|_2 < \varepsilon. \quad (27)$$

Thus, along a single scan path, the ε -effective receptive field of the Mamba prompter is concentrated within a local temporal neighborhood:

$$\mathcal{R}_{\text{eff},t}^\varepsilon(\text{Mamba}) \subseteq \{\tau : |t - \tau| \leq \Delta_\varepsilon\}. \quad (28)$$

Under the proposed CSM, image patches are traversed through multiple cyclic scan paths. Let $d_p(t, \tau)$ denote the path distance between token τ and token t under the p -th scan path. Then the effective receptive field can be described more generally as

$$\mathcal{R}_{\text{eff},t}^\varepsilon(\text{Mamba}) \subseteq \bigcup_{p=1}^P \{\tau : d_p(t, \tau) \leq \Delta_\varepsilon\}, \quad (29)$$

where P is the number of scan paths.

Equation (29) shows that multi-path scanning enlarges the spatial coverage and revisits local regions from different directions. Therefore, after aggregation across multiple scan paths, the Mamba prompter still exhibit broad spatial coverage in a theoretical sense, while its effective influence remains locality-biased and path-dependent. This explains why Mamba is particularly sensitive to locally discriminative information.

Complementarity in MpVH. The above analysis suggests that ViT and Mamba offer complementary feature sensitivities:

- ViT provides globally aggregated contextual semantics, which is beneficial for preserving class-level structure and semantic coherence.
- the Mamba prompter provides locality-biased discriminative information, which is beneficial for emphasizing fine-grained and spatially informative details.

Let C_{global} denote the ViT features and $\mathcal{F}_{\text{local}}$ denote the adapted prompts generated from the Mamba branch. In MpVH, these two components interact through the prompting mechanism and the

MoEA adapter to form an enhanced descriptor

$$x' = \phi(C_{\text{global}}, \mathcal{F}_{\text{local}}), \quad (30)$$

where $\phi(\cdot, \cdot)$ denotes the feature interaction implemented by the prompt injection and adaptation modules.

Under a fixed hash length, descriptors that preserve both global semantic consistency and local discriminability are better able to retain critical information under compression and quantization. Accordingly, MpVH employs Mamba-generated prompts to guide ViT in learning importance-aware image descriptors, thereby alleviating the loss of discriminative information during hash encoding.